

Data-driven adjoint-based algorithms for estimation, learning, and control

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Idea and main result

Application to an example system

Data-driven estimation: behind the algorithm

Data-driven estimation: experimental case study

Overview

Data-driven control

- ▶ highly popular
- ▶ this IFAC: Data-Driven Control Theory I, II, III, ...
- ▶ many approaches based on Willems' fundamental lemma
 - ▶ e.g., DeePC
 - ▶ non-parametric representation implicitly encoded in the data
 - ▶ see WeC06.3 for connections to subspace predictive control

Today: data-driven adjoint-based algorithms

- ▶ do **not** implicitly estimate a model
- ▶ only very mild assumptions (LTI)
- ▶ adjoints are at the heart of the field
 - ▶ role in optimal control, minimum energy control, and optimal ILC
 - ▶ key in frequency-domain ILC (`filtfilt`)
 - ▶ mathematical foundation for duality controllability and observability

Some well-known algorithm where we use models

1. Iterative Learning Control (ILC) $f_{j+1} = f_j + Le_j$,
with L typically based on a model
2. Uncertainty estimation for robust control $y = \Delta f$,
where a model $\hat{\Delta}$ is of Δ is made, and $\|\hat{\Delta}\|_{\infty}$ is computed

Today: unified framework for data-driven counterpart

- ▶ adjoint-based for estimation, learning, and control
 - ▶ optimal ILC
 - ▶ system norms
 - ▶ system properties
 - ▶ ...
- ▶ many extensions, multivariable, resetting, etc.

Two algorithms in more detail

ILC

criterion:

$$\mathcal{J}(f_{j+1}) = \|e_{j+1}\|_{w_e} + \|f_{j+1}\|_{w_f}$$

is minimized by iteration:

$$f_{j+1} = (I - \epsilon_j W_f) f_j + \epsilon_j J^T W_e e_j$$

$\mathcal{H}_\infty$

estimator:

$$\beta_j = \sqrt{\frac{z_j^T f_j}{f_j^T f_j}}$$

converges to $\mathcal{H}_\infty$ norm via iteration:

$$f_{j+1} = \epsilon_j J^T J f_j$$

Key observation: involves J^T

On J^T

$$\blacktriangleright \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{N-1} \end{bmatrix}}_y = \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_J \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{bmatrix}}_u$$

► traditional model-based approaches: make model $\hat{J}$ of J , then take $\hat{J}^T$

► the key idea

$$\underbrace{\begin{bmatrix} h_0 & h_1 & \cdots & h_{N-1} \\ 0 & h_0 & \ddots & h_{N-2} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & h_0 \end{bmatrix}}_{J^T} = \underbrace{\begin{bmatrix} 0 & \cdots & 0 & 1 \\ \vdots & \ddots & \ddots & 0 \\ 0 & \ddots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \end{bmatrix}}_{\mathcal{T}} \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_J \underbrace{\begin{bmatrix} 0 & \cdots & 0 & 1 \\ \vdots & \ddots & \ddots & 0 \\ 0 & \ddots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \end{bmatrix}}_{\mathcal{T}}$$

► evaluate J^T by: time reversal $\mathcal{T} \Rightarrow$ filter through $J \Rightarrow$ time reversal $\mathcal{T}$

► J^T is the adjoint operator (recall `filtfilt` in ILC)

Idea and main result

Application to an example system

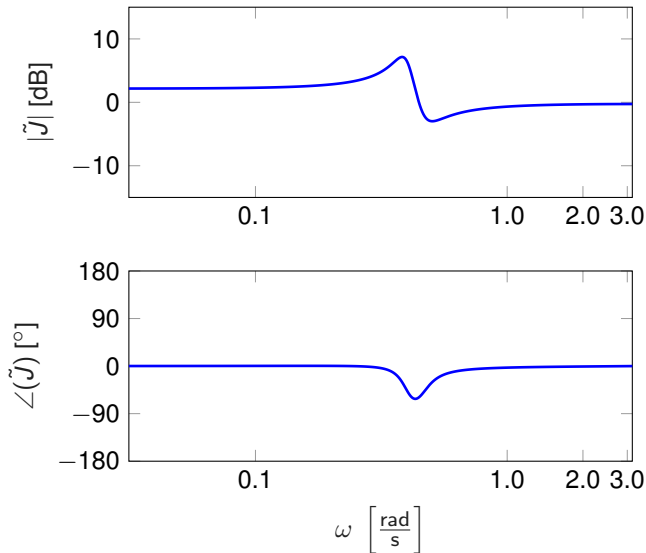
Data-driven estimation: behind the algorithm

Data-driven estimation: experimental case study

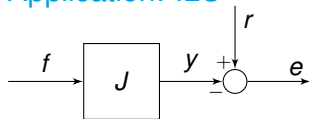
Overview

Example system

- ▶ transfer function $\tilde{J}$
- ▶ convolution matrix J with $N = 100$
- ▶ benchmark system (see extended paper for many use cases)



Application: ILC

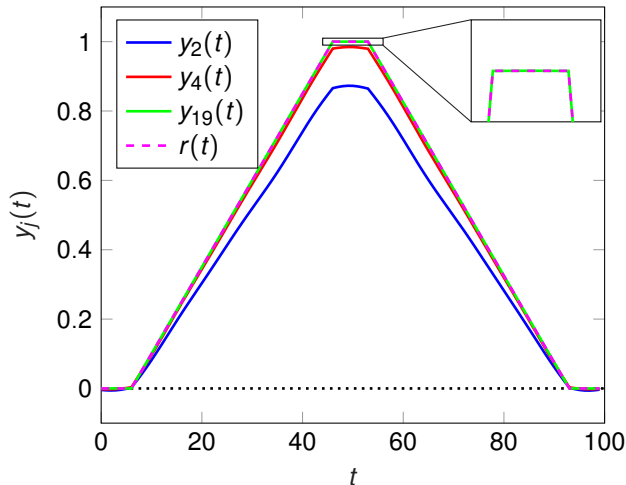


► criterion

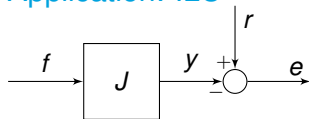
$$\mathcal{J}(f_{j+1}) = \|e_{j+1}\|_2$$

► minimized by iteration

$$f_{j+1} = f_j + \epsilon_j J^T W_e e_j$$



Application: ILC

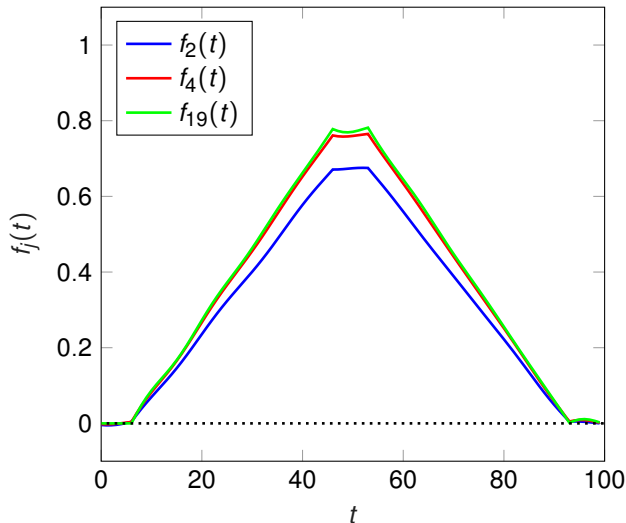


- criterion

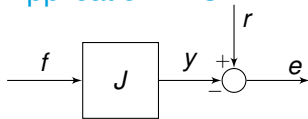
$$\mathcal{J}(f_{j+1}) = \|e_{j+1}\|_2$$

- minimized by iteration

$$f_{j+1} = f_j + \epsilon_j J^T W_e e_j$$



Application: ILC

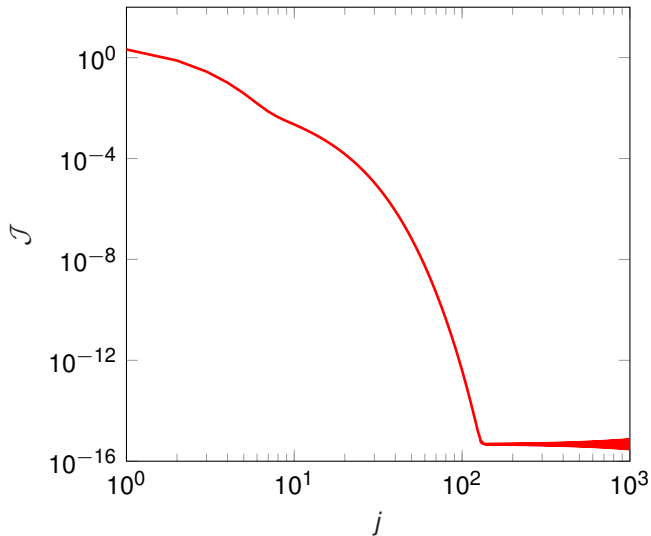


- criterion

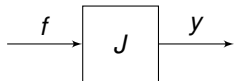
$$\mathcal{J}(f_{j+1}) = \|e_{j+1}\|_2$$

- minimized by iteration

$$f_{j+1} = f_j + \epsilon_j J^T W_e e_j$$



Application: $\mathcal{H}_\infty$

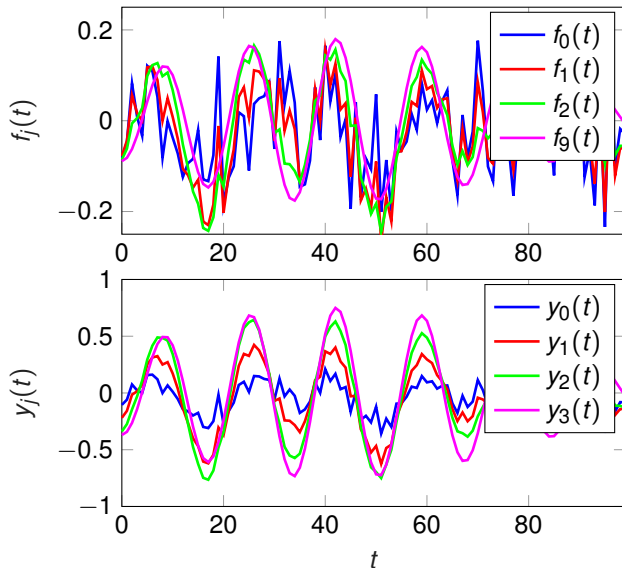


► estimator

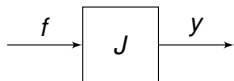
$$\beta_j = \sqrt{\frac{z_j^T f_j}{f_j^T f_j}}$$

► with iteration:

$$f_{j+1} = \epsilon_j J^T J f_j$$



Application: $\mathcal{H}_\infty$



- estimator

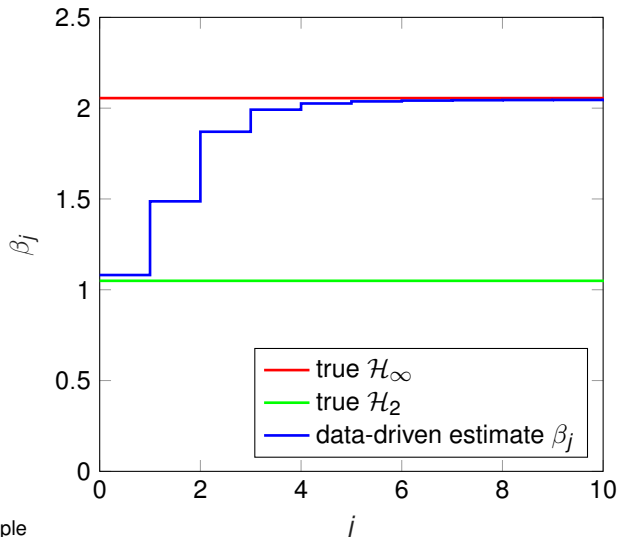
$$\beta_j = \sqrt{\frac{z_j^T f_j}{f_j^T f_j}}$$

- with iteration:

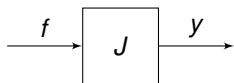
$$f_{j+1} = \epsilon_j J^T J f_j$$

- Observations

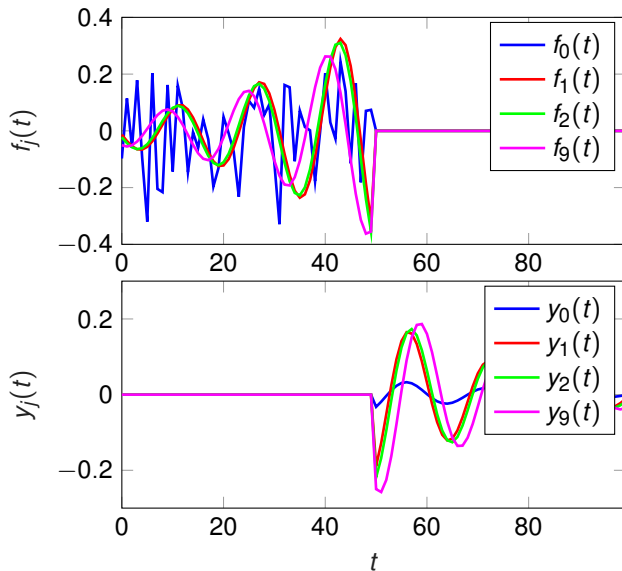
- $\beta_j \rightarrow \|J\|_\infty$, for $N \rightarrow \infty$
- $\beta_0 \approx \|J\|_2$ (proof available)
- we can do much more - next example



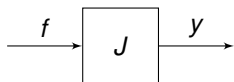
Application: Hankel norm



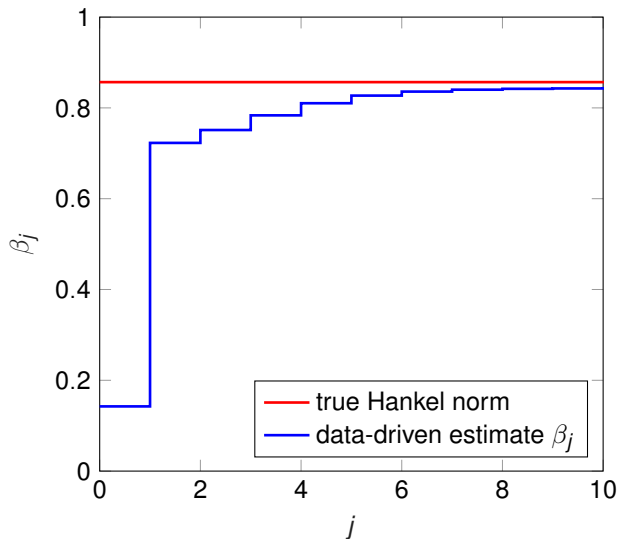
- ▶ accurate data-driven Hankel norm estimation
- ▶ interpretation Hankel norm: induced norm of past inputs to future outputs



Application: Hankel norm



- ▶ accurate data-driven Hankel norm estimation
- ▶ interpretation Hankel norm: induced norm of past inputs to future outputs
- ▶ many more system properties (see forthcoming paper)



Idea and main result

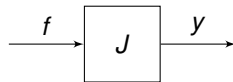
Application to an example system

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Overview

How to estimate norms of J without modeling?



▶ infinite time $\tilde{J}$: $\|\tilde{J}\|_\infty = \sup_{f \in \ell_2} \frac{\|y\|_2}{\|f\|_2}$

▶ finite time J :

▶ experiment length N :

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}}_{=y} = \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_{=J} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix}}_{=f}$$

▶ then: $\sup_{f \in \ell_2} \frac{\|y\|_2}{\|f\|_2} = \sup_{f \in \ell_2} \sqrt{\frac{y^T y}{f^T f}} = \sup_{f \in \ell_2} \sqrt{\frac{f^T J^T J f}{f^T f}}$

▶ maximal if f corresponds to the eigenvector $\bar{f}$ of $\lambda^{\max}(J^T J)$:

$$\sup_{f \in \ell_2} \frac{\|y\|_2}{\|f\|_2} = \sqrt{\frac{\bar{f}^T J^T J \bar{f}}{\bar{f}^T \bar{f}}} = \sqrt{\lambda^{\max}(J^T J)}$$

$\Rightarrow$ connection $\|G\|_\infty$ and $\lambda^{\max}(J^T J)$

Analysis of $J^T J$

- ▶ as before: exploit Toeplitz structure

$$J^T = \begin{bmatrix} h_0 & h_1 & \cdots & h_{N-1} \\ 0 & h_0 & \cdots & h_{N-2} \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & h_0 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{=\mathcal{T} \text{ (time reversal)}} \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \cdots & 0 \\ \vdots & & \ddots & \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_{=J} \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{=\mathcal{T}}$$

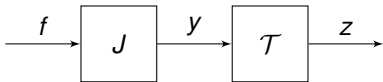
- ▶ hence: $J^T J = (\mathcal{T}J)(\mathcal{T}J)$

- ▶ Hankel structure: $\mathcal{T}J = (\mathcal{T}J)^T \Rightarrow |\lambda^{\max}(\mathcal{T}J)| = \sqrt{\lambda^{\max}(J^T J)}$

- ▶ thus: $|\lambda^{\max}(\mathcal{T}J)| \rightarrow \|G\|_{\infty}$ for $N \rightarrow \infty$

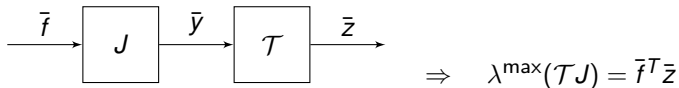
- ▶ remark: $\lambda^{\max}(J) = h_0$

Resulting matrix $\mathcal{T}J$:



We have: $|\lambda^{\max}(\mathcal{T}J)| \rightarrow \|\tilde{J}\|_{\infty}$ for $N \rightarrow \infty$

- ▶ Given $\bar{f}$ of $\lambda^{\max}(\mathcal{T}J)$, with $\|\bar{f}\|_2 = 1$. Evaluate:



- ▶ **Key problem:** how to obtain $\bar{f}$?

$\Rightarrow$ **power iteration (linear algebra):** $x_{j+1} = \frac{Ax_j}{\|Ax_j\|}$, then $\lambda_k = x_j^T Ax_j$ converges to $\lambda^{\max}(A)$

Result: $f_j \rightarrow \bar{f}$ for $j \rightarrow \infty$ (global convergence)

Iteratively learning $\mathcal{H}_{\infty}$ -norm (SISO)

- ▶ replace matrix ('model') J by evaluation on the true system
- ▶ then, $\beta_j = |f_j^T z_j| \rightarrow \|G\|_{\infty}$ for $j \rightarrow \infty$

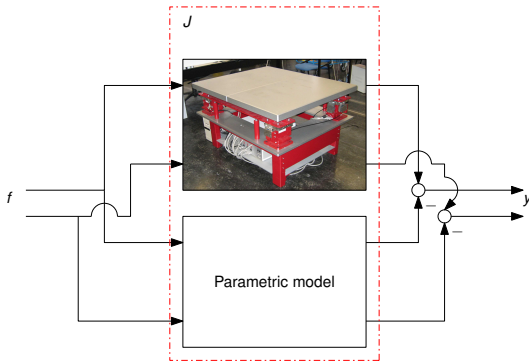
Idea and main result

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Data-driven estimation: behind the algorithm

Data-driven estimation: experimental case study

Overview



Motivation: robust control

- ▶ model uncertainty bound γ
- ▶ if $\|J\|_{\infty} \leq \gamma$: **guarantees!**
- ▶ if $\|J\|_{\infty} > \gamma$: **no guarantees**
- ▶ if $\|J\|_{\infty} \ll \gamma$: **conservative**

Need accurate bound $\|J\|_{\infty}$

Frequency domain robust control

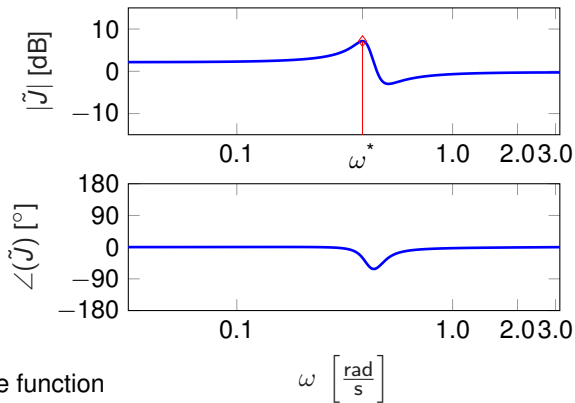
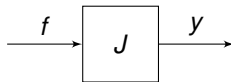
- ▶ J SISO, LTI, discrete time stable
frequency domain interpretation:

$$\|\tilde{J}\|_{\infty} = \sup_{\omega} |\tilde{J}(e^{j\omega})| = |\tilde{J}(e^{j\omega^*})|$$

- ▶ J MIMO, LTI, discrete time stable
frequency domain interpretation:

$$\|\tilde{J}\|_{\infty} = \sup_{\omega} \bar{\sigma}(\tilde{J}(e^{j\omega}))$$

data-driven approach in (Oomen et al.2014)

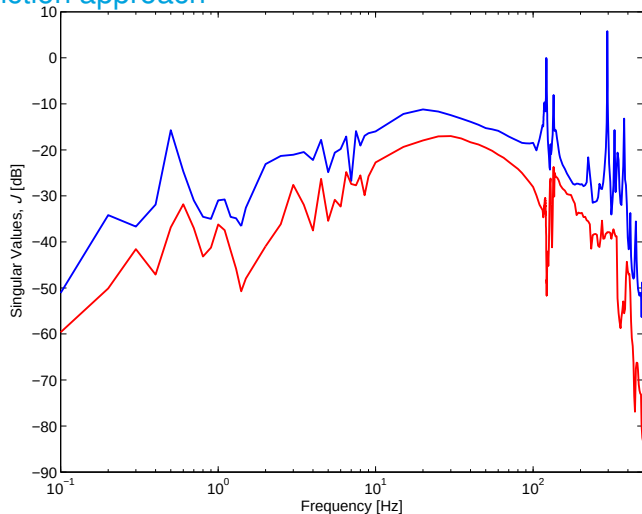


Classical model-based approaches

- ▶ identify parametric model $\hat{J}(\theta)$
 $\Rightarrow$ compute $\|J(\theta)\|_{\infty} \Rightarrow$ **model quality?**
- ▶ identify non-parametric frequency response function
 $\Rightarrow$ **intergrid error**

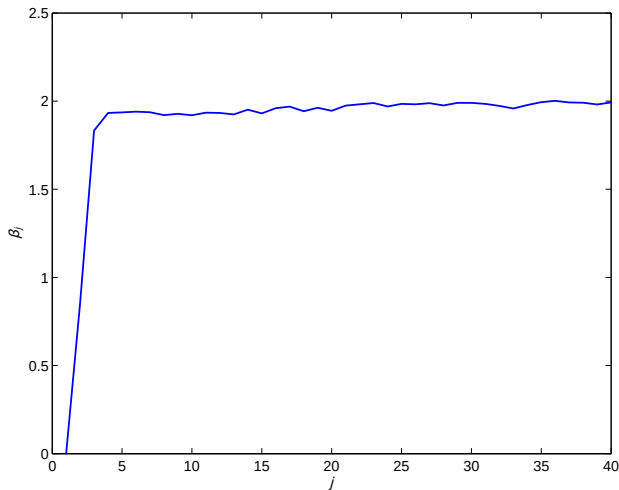
Traditional frequency response function approach

- ▶ Nonparametric identification of J
- ▶ long measurement
- ▶ $\|J\|_{\infty, FRF} = 1.95$
- ▶ $\omega^* \approx 296$ Hz

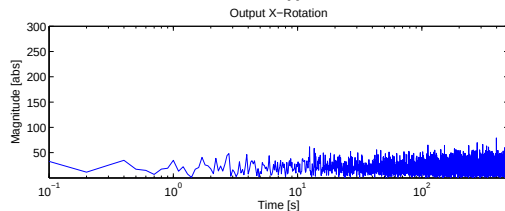
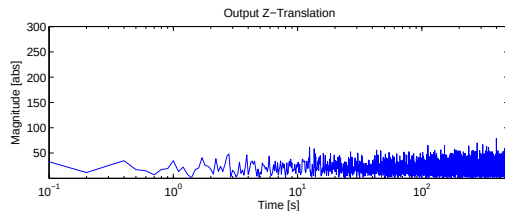
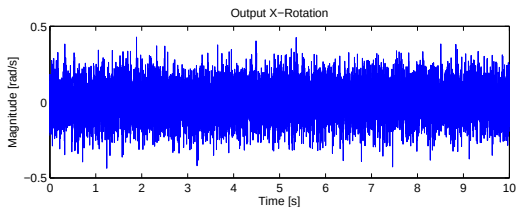
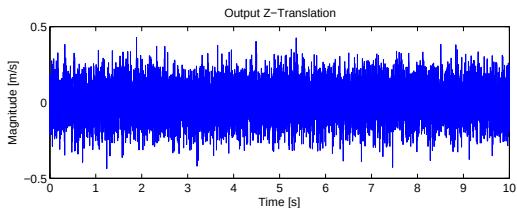


Data-driven algorithm

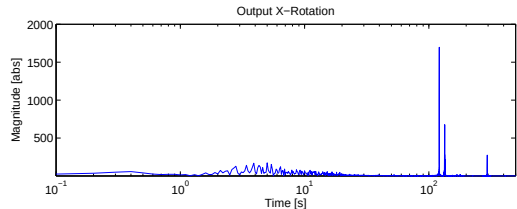
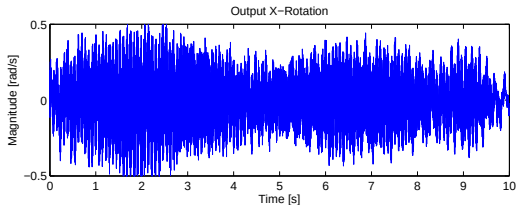
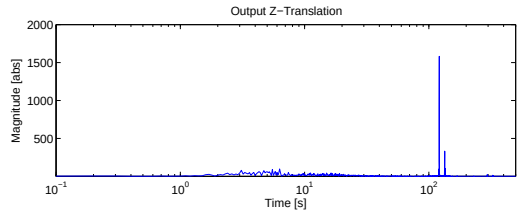
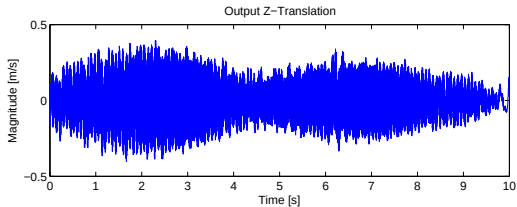
- ▶ $\|J\|_{\infty}^{\text{data-driven}} = 1.997$
- ▶ $(\|J\|_{\infty}^{FRF} = 1.95)$



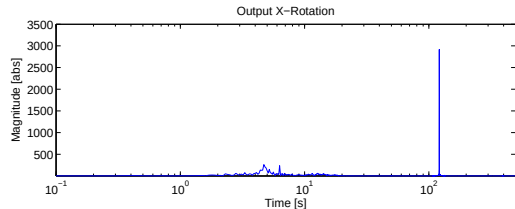
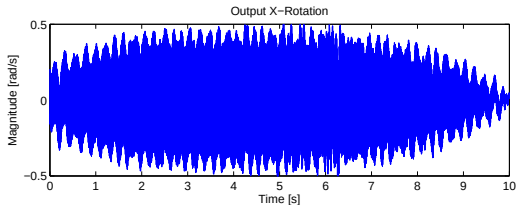
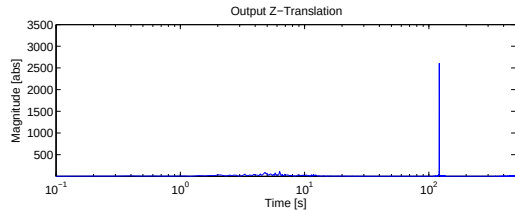
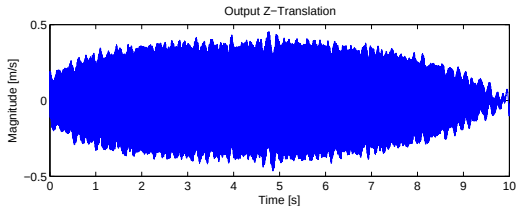
- Initial input: white noise

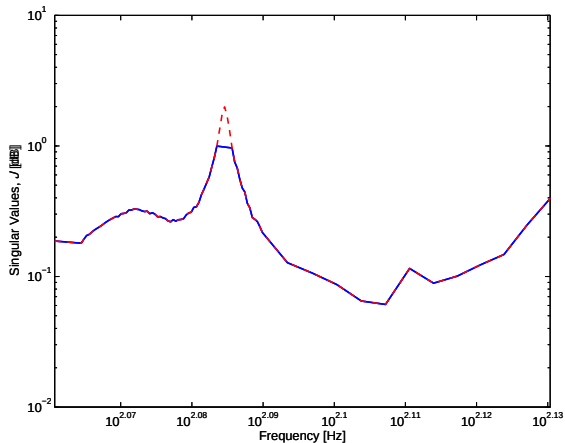
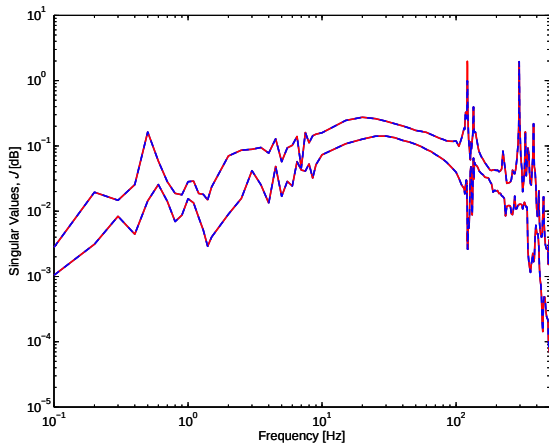


- Output after 2 iterations: dominant frequency components



- Output after 40 iterations: single component at $\omega^* = 121$ Hz
- $\omega_{FRF}^* \neq \omega_{\text{data-driven}}^*$
(121 Hz $\neq$ 296 Hz)





- ▶ **initial long measurement:** baseline, different frequency?
- ▶ **much longer validation measurement:** we missed a peak!
- ▶ **iterative approach:** useful for experiment design

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Conclusion

- ▶ unified overview of data-driven adjoint estimation algorithms

Extensions: a unified and broad framework!

- ▶ multivariable (Oomen et al.2014) (Bolder et al.2018) (Aarnoudse & Oomen2021)
- ▶ reset vs. continuous operation (Oomen & Rojas2024)
- ▶ noise analysis (Rojas et al.2012)
- ▶ connections to VRFT (Formentin et al.2015) IFT (Aarnoudse et al.2025)
- ▶ many many extensions, other properties, etc.: **paper in preparation**

- Aarnoudse, L., den Toom, P. & Oomen, T. (2025), 'Randomized iterative feedback tuning for fast MIMO feedback design of a mechatronic system', *Control Engineering Practice* **154**, 106152.
- Aarnoudse, L. & Oomen, T. (2021), 'Model-free learning for massive MIMO systems: Stochastic approximation adjoint iterative learning control', *IEEE Control Systems Letters (L-CSS)* **5**(6), 1946–1951.
- Bolder, J., Kleinendorst, S. & Oomen, T. (2018), 'Data-driven multivariable ILC: Enhanced performance by eliminating L and Q filters', *International Journal of Robust and Nonlinear Control* **28**(12), 3728–3751.
- Formentin, S., Bisoffi, A. & Oomen, T. (2015), Asymptotically exact direct data-driven multivariable controller tuning, in '2015 IFAC Symposium on System Identification', Beijing, China, pp. 1349–1354.
- Oomen, T. & Rojas, C. R. (2024), 'Reset-free data-driven gain estimation: Power iteration using reversed-circulant matrices', *Automatica* **161**, 111505.
- Oomen, T., van der Maas, R., Rojas, C. R. & Hjalmarsson, H. (2014), 'Iterative data-driven $\mathcal{H}_\infty$ norm estimation of multivariable systems with application to robust active vibration isolation', *IEEE Transactions on Control Systems Technology* **22**(6), 2247–2260.
- Rojas, C. R., Oomen, T., Hjalmarsson, H. & Wahlberg, B. (2012), 'Analyzing iterations in identification with application to nonparametric $\mathcal{H}_\infty$ -norm estimation', *Automatica* **48**(11), 2776–2790.